

SpiralMap: A Python library of the Milky Way's spiral arms

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Software

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Summary

Mapping the structure of the Galaxy has been an ongoing endeavour for several decades, thanks to which we have come to piece together the components it is made up of such as the discs/bulge/halo and so on (Bland-Hawthorn & Gerhard, 2016). Of particular interest is also the signature of non-axisymmetry in the disc, principally present in the form of the central Galactic bar, the warp, and various spiral like features fanning across a large portion of the disc. Over the years, various groups have surveyed the Galaxy across wavelengths (Radio to optical) and have deduced the rich variety of spiral structure present in the disc. While some of the papers in literature provide machine readable data to trace out the spiral arms in their model, often this can be a cumbersome exercise for a user simply interested in extracting the coordinates and/or overplotting the spiral arms on another plot of interest, such as while comparing the locations of the arms to features in the velocity field (Khanna et al., 2023; Poggio et al., 2025). With SpiralMap we present a library of the major spiral arm models (and maps) of the Galaxy.

Statement of need

The SpiralMap package is written in Python, and allows the user to both extract the 2D trace and overplot the spiral arms in cartesian/polar coordinates in both Heliocentric (HC) and Galactocentric (GC) frames. A summary of the models currently included is provided in the Table below, where we have tried to include models from across the electromagnetic spectrum, and based on various tracers (gas/stars etc.). Other models can easily be included on request. In the near future, we anticipate the availability of 3D spiral arm traces for the Galaxy in literature which can also be included in SpiralMap.

Table with 2 columns: Model, Description. Rows include Taylor_Cordes_1992, Vallee_1995, Drimmel_NIR_2000, Levine_2006, Hou_Han_2014, Reid_2019, Poggio_2021, Gaia_2022, Drimmel_Ceph_2024.

28 A few example plots that can be generated using the package are included below. For more
29 details, we point to the following links: a) Documentation: [readthedocs](#), b) Demonstration:
30 [Jupyter notebook](#), and c) [Github repository](#).

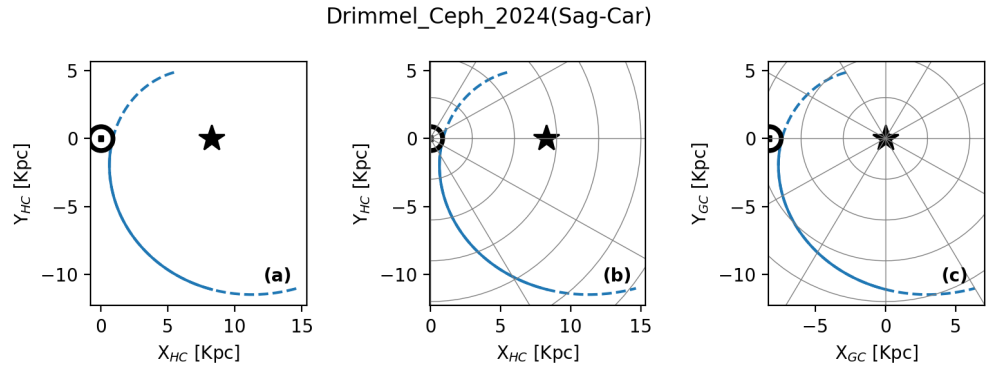


Figure 1: Cartesian projection of the Drimmel_Ceph_2024 model shown for a particular arm (Sag-Car). We show this arm in HC (a), HC with a polar grid in the background (b), and in GC frame with a polar grid in the background (c).

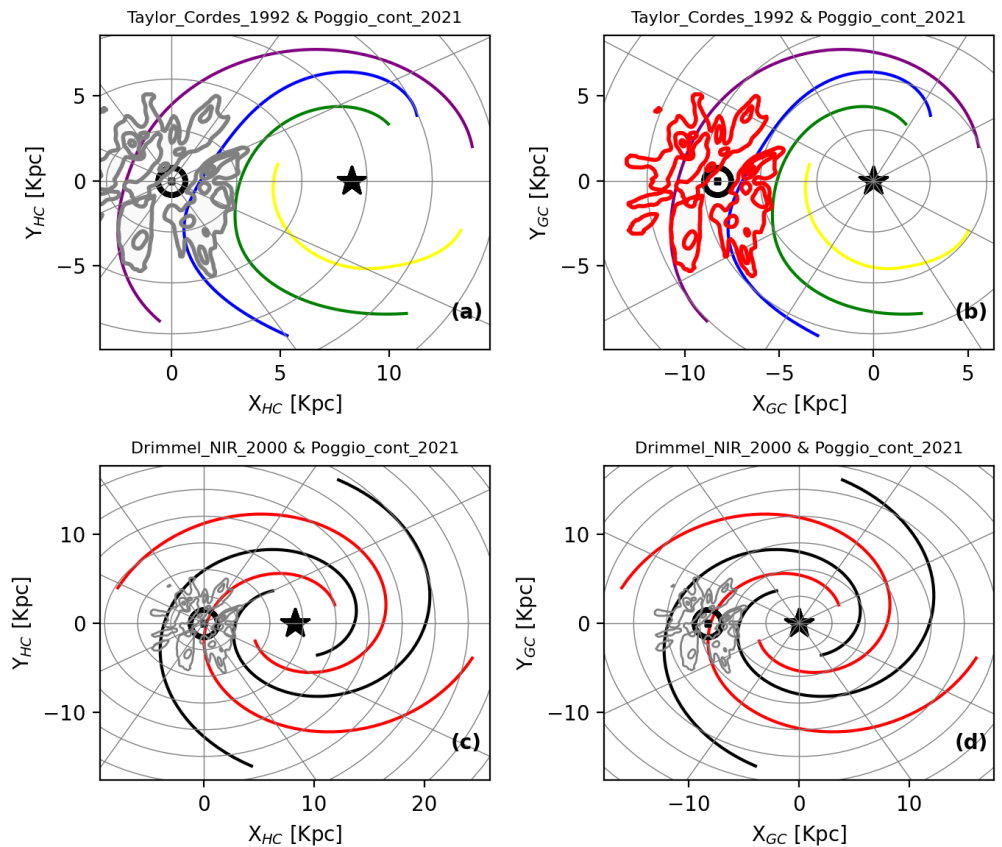


Figure 2: Cartesian projections of multiple models plotted together with a polar grid in the background. We show the Taylor_Cordes_1992 & Poggio_2021 models in HC (a) and GC (b) frames, and similarly, the Drimmel_NIR_2000 & Poggio_2021 models in HC (c) and GC (d) frames.

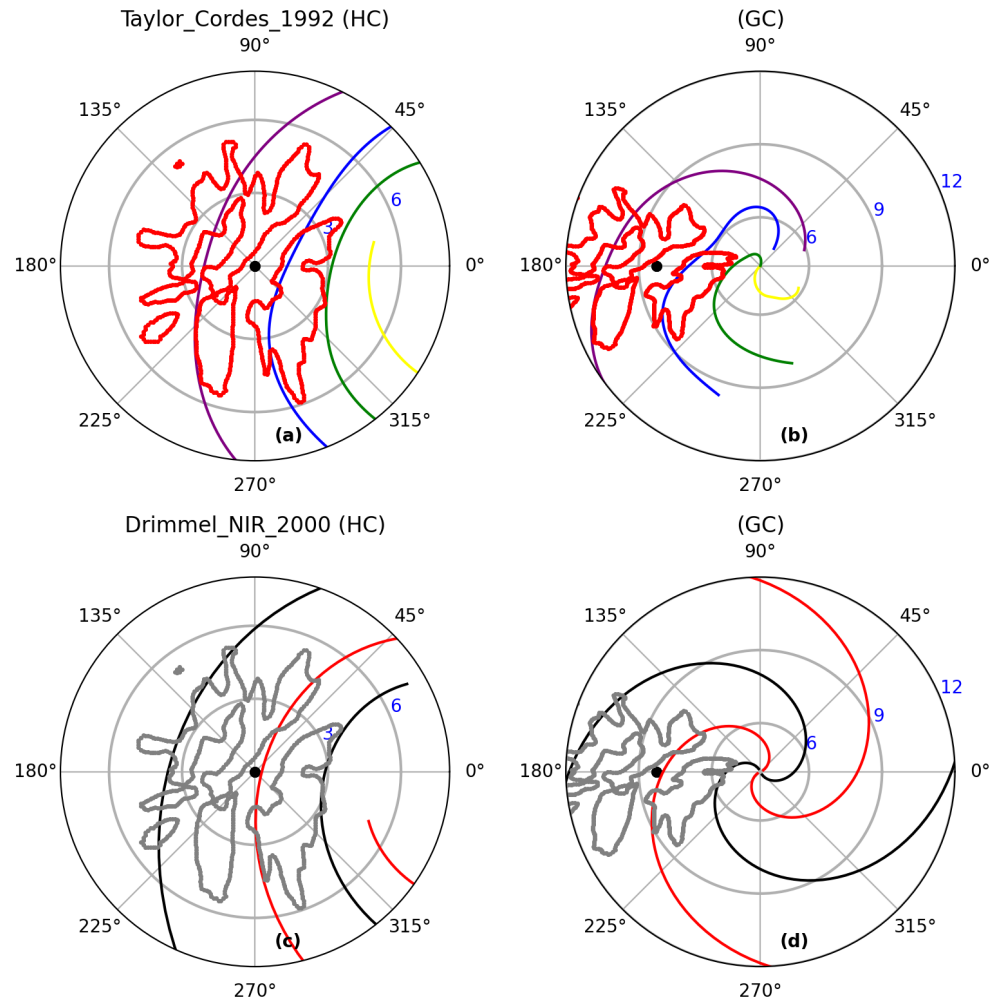


Figure 3: Polar projections of multiple models plotted together. We show the Taylor_Cordes_1992 & Poggio_2021 models in HC (a) and GC (b) frames, and similarly the Drimmel_NIR_2000 & Poggio_2021 models in HC (c) and GC (d) frames.

State of the field

To the best of our knowledge, there is no other publicly available package providing similar tools to the community. We acknowledge the individual surveys and papers that have mapped out the spirality of the Milky Way, however we have also observed that it is quite easy to miss out on historically derived models, likely because there is no standard way of publishing a spiral arm model. Some papers provide datafiles, others provide large tables which still require individual users to extract the data, and doing so for several papers can be a cumbersome exercise. With our package having all the major models under one umbrella interface allows for an easier comparison between the various tracers (stars, gas, etc) of such non-axisymmetry, and also facilitates fair visibility of the various efforts carried out over decades. Indeed with SpiralMap we intend to address this gap, by providing a user-friendly interface encompassing all the major spiral arm models and maps currently offering traces in 2D, and very soon to extend this to 3D.

44 Software Design

45 The main motivation behind SpiralMap is to provide a Python3 based user-friendly, object-
46 oriented API that allows users to generate spatial traces of the various Spiral arm maps and
47 models of the Milky Way, available in literature. The package relies on a minimal set of
48 dependencies, including on well maintained community tools such as Astropy. While most
49 of the models currently included in the package are analytical, i.e. the code generates the
50 spiral trace for the user, in a few select cases these are read from pre-generated files, which
51 are provided in user-friendly formats such as pickle or FITS files. The individual spiral arm
52 models are all available in literature but in disparate forms (datafiles, analytical descriptions,
53 etc). Our aim here is to provide the user with a one-stop python class where the application
54 of the various spiral models can be homogenised, in particular accessing both heliocentric
55 and galactocentric coordinates of the spiral tracers, the ability to quickly plot both cartesian
56 or polar projections, or to select which (or all) spiral arms to plot for a particular model etc.
57 Additionally, the user still has access to the raw forms of the spiral models as were presented
58 in literature, except here in a machine readable form.

59 Research Impact Statement

60 SpiralMap has been available to the community since June 2025. PyPI stats shows that the
61 package has been downloaded about 1500 times since release, averaging about 200 downloads
62 per month. The supporting document on the arXiv has also been cited by two peer reviewed
63 papers (ex: Trentin et al. 2025) as of Feb 2026, with at least 3 other papers in prep that have
64 requested particular model updates. To this end, we pushed a new version in February 2026
65 with additional spiral arm models appended to the package, and expect another update in
66 March 2026 which will include the first 3D spiral arm model.

67 AI usage disclosure

68 No generative AI tools were used in the development of this software, writing of this manuscript,
69 or the preparation of supporting materials.

70 Example Scientific application

71 As an example of a science case, we reproduce figures from Khanna et al. (2025) (hereafter
72 K24) where SpiralMap was used. In particular, K24 constructed a model for the stellar density
73 distribution in Red Clump stars in the Milky Way, and then compared the residuals of their
74 best-fit models with the locations of non-axisymmetric structures such as spiral arms. The
75 figures below show two of their residual plots overlaid with spiral models (Drimmel_NIR_2000,
76 Drimmel_Ceph_2024, & Reid_2019) using SpiralMap.

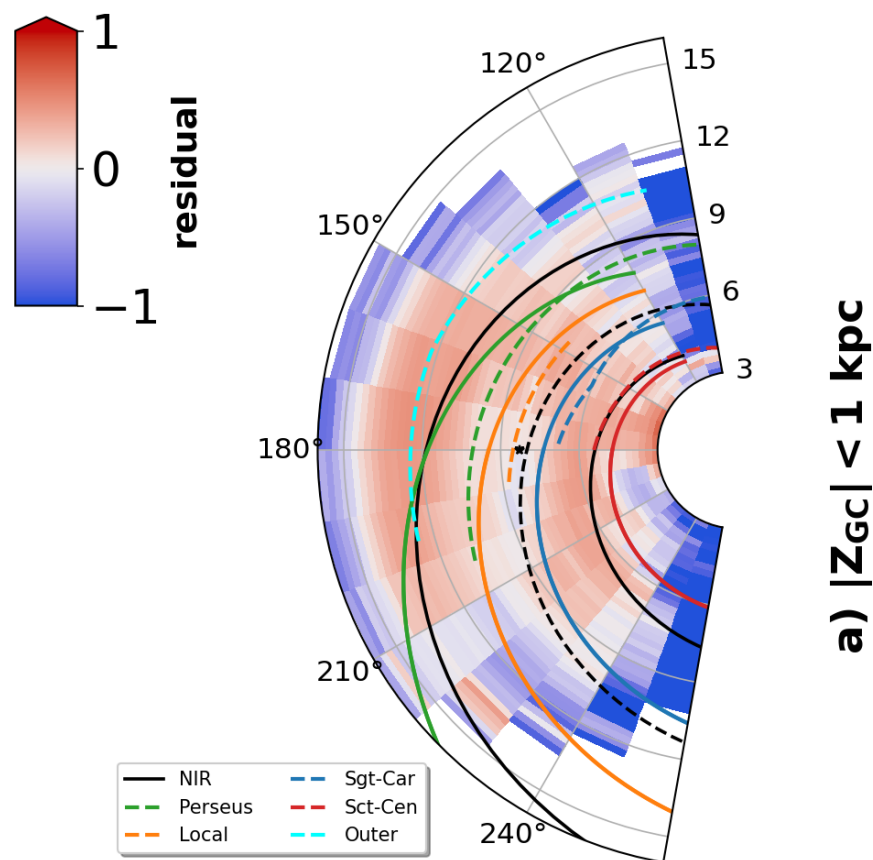


Figure 4: An example science case where SpiralMap can be useful. Here we show two figures reproduced from K24 (with permission), comparing the residuals between a best-fit model and data, and the locations of spiral arm models in the Milky Way. This figure shows the Drimmel_NIR_2000, Drimmel_Ceph_2024, & Reid_2019 overplotted together, shown in polar projection.

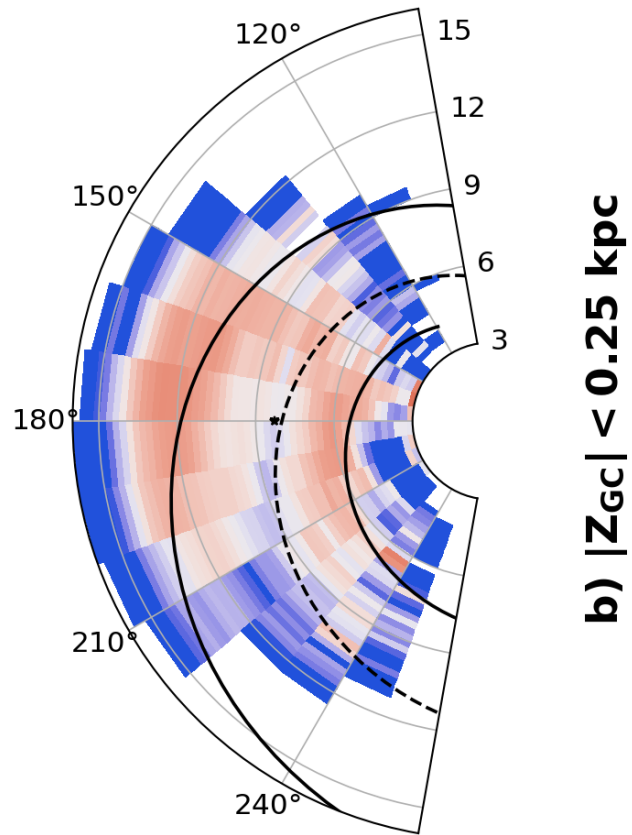


Figure 5: Same as above, but showing only the Drimmel_NIR_2000 model.

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77 A very basic example of using SpiralMap is shown below, where we access all information
78 about one particular arm ( Sag-Car) in one particular model Drimmel_Ceph_2024,
79 #####
80 ##### Readout a single arm from a single model ###
81 #####
82 import SpiralMap as sp
83 from SpiralMap import polar_style
84
85 Rsun=8.277
86 spirals = sp.main_(Rsun=Rsun)
87 use_model = `Drimmel_Ceph_2024`
88 spirals.getinfo(model=use_model)
89 plotattrs = {'plot':False}
90 spirals.readout(plotattrs,model=use_model,arm=`Sag-Car`)
91
92 #####
93 More common examples are shown in the accompanying Jupyter notebook.

```

Availability

The source code for SpiralMap is available on [GitHub](#), and the full documentation is hosted on [Read the docs](#).

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